

ST-MG Prompt

Overall framework · train-only priors

Historical input

① **Dual Temporal Semantics**
same 144-step grid

Fine

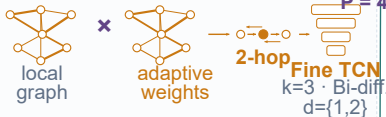
local variation
causal local context

Coarse

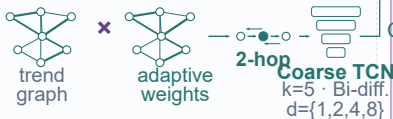
long-range trend
 $w = \{6, 18, 36\}$
Fine / Coarse = $1/2 + 1/2$

② **Graph-Temporal Encoding**
two semantic views

Micro
 $\Delta W_{\text{spd}} + \text{distance}$



Macro
Patv trend + distance



③ **Cross-Granularity Fusion**

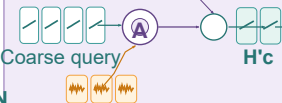
Macro-trend prompt

P1 P2 P3 P4

P = 4 tokens / turbine



Recent Fine → Coarse



Recent Fine = 24

$H'f = \text{LN}(Hf + G \cdot O_{m \rightarrow f})$
 $H'c = \text{LN}(Hc + (1-G) \cdot O_{f \rightarrow c})$

④ **Forecasting Context**

last + macro mean
context vector

Node-Horizon ST Prompt

Direct decoder · H=10

future horizon
Y-hat in $R^{(B \times 10 \times 134)}$
H3 H6 H10

Training only · MS-MG-DWU

Y-hat Y + mask

H3 / H6 / H10
prefix difficulty

site difficulty
dynamic weights

MS-MG-DWU

objective only

optimization branch; no forward-path input